Title: Predicting High-Risk Drug Interactions in FAERS Data

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Important Notes:

- Each group must submit ONLY one copy by the single team leader!
- Do not worry! Since you will list all team members in the table above.

1. Introduction

- Background**:** Adverse Drug Events (ADEs) are a leading cause of hospitalization and mortality worldwide. The FDA Adverse Event Reporting System (FAERS) collects millions of reports to monitor drug safety post-market.
- Motivations: As a team, we want to move beyond simple data exploration to identify hidden patterns in drug co-prescription that lead to severe clinical outcomes. This project is motivated by the need for automated systems that can alert healthcare providers about dangerous drug-drug interactions (DDIs) that are not yet documented in clinical trials.

2. Data Sets

Domain: Healthcare and Pharmacovigilance.
Source: openFDA / FAERS API (2024-2025 quarterly data).
Link: https://open.fda.gov/apis/drug/event/

Size: The available dataset contains 333,446 total records divided in 1710 reports (https://open.fda.gov/apis/drug/event/download/) but we will only take some reports from the year 2025.
Proposed Sample Size: We will process a subset of at least 120,000 to 150,000 rows to ensure it meets the course requirement (>100k).

Variables: 52 fields available per report, focusing on 10 key features for our research:

- Key features: include patientonsetage , patientsex , medicinalproduct, and route.
- Target Variable (Classification): Clinical Severity (derived from seriousness indicators like seriousnessdeath, seriousnesshospitalization, and seriousnessdisabling).
- Association Items: medicinalproduct and reactionmeddrapt (Specific reaction).

3. Research Problems

- Interaction Mining: Identify which specific combinations of drugs (e.g., Drug A + Drug B) have the strongest statistical association with life-threatening reactions.
- Severity Prediction: Build a classification model to predict the "Severity Level" of a report based on the patient's profile and medication history.
- Challenges: The data is semi-structured (JSON) and highly imbalanced (severe cases are rarer than mild ones). We must handle nested lists of drugs and reactions efficiently.

4. Potential Solutions

- Problem 1 (Unsupervised): We will implement Association Rule Mining. We will compare the performance and rules generated by the Apriori algorithm versus FP-Growth to find high-lift drug combinations.
- Problem 2 (Supervised): We will implement a classification pipeline. We will compare an Ensemble Classifier (Random Forest), against Logistic Regression or SVM.

5. Evaluations

Approach: We will use a 70/30 Hold-out evaluation for classification tasks and Cross-Validation to ensure model stability.

Metrics:

- For Association Rules: Support, Confidence, and Lift (to filter non-random associations).
- For Classification: Since medical data is often imbalanced, we will prioritize F1-Score and AUC-PR over simple Accuracy.

6. Expected Outcomes

- A ranked list of the most dangerous drug-drug interactions found in the 2024-2025 data.
- A comparative analysis showing which machine learning model best predicts patient outcome severity.
- A Python-based pipeline capable of processing raw JSON medical reports into actionable safety insights.